

PROGRAMMING METHODOLOGY

SE1012

LAB 02 – REPORT

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Activity 1: Displaying Text in C

Activity 1

Write a C program to display:

1. "Hello! Welcome to SLIIT"
2. "First line\nThis is the second line"

3.

```
  X
X  X
X  X
X  X
X  X
X  X
X  X
X  X
X
```

Instructions

1. Open the Terminal using the Applications Menu or the keyboard shortcut (Ctrl + Alt + T).
2. Navigate to the Desktop.
3. Create a working directory with your IT number.
4. Inside the directory, create a new C file named `display_text.c`.
5. Write the C program as described in Problem 1.
6. Compile and run the program to ensure it works correctly.

Code:

PM-2-LAB / lab-02-2026-it26100408 / lab-02-2026-it26100408 / hello_world.c

 it26100408 Completed Lab Activity 1

Code Blame 7 lines (6 loc) · 143 Bytes

```
1  #include <stdio.h>
2
3  int main() {
4      printf("Hello! Welcome to SLIIT\n");
5      printf("First line\nThis is the second line\n");
6      return 0;
7  }
```

PM-2-LAB / lab-02-2026-it26100408 / lab-02-2026-it26100408 / problem3 /



it26100408 Completed Problem 3: Diamond pattern

Code

Blame

15 lines (13 loc) · 312 Bytes

```
1  #include <stdio.h>
2
3  int main(){
4
5      printf("  x  \n");
6      printf(" x x  \n");
7      printf("x  x  \n");
8      printf(" x x  \n");
9      printf(" x  x  \n");
10     printf("  x x  \n");
11     printf(" x  x  \n");
12     printf("  x x  \n");
13     printf("   x  \n");
14     return 0;
15 }
```

Out put:

```
vihangam@UbuntuVM: ~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408
bash: cd: lab-02-2026-it26100408: No such file or directory
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB$ mkdir lab-02-2026-it26100408
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB$ cd lab-02-2026-it26100408
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408$ git clone https://github.com/it26100408/PM-2-LAB.git
Cloning into 'PM-2-LAB'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408$ cd lab-02-2026-it26100408
bash: cd: lab-02-2026-it26100408: No such file or directory
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408$ mkdir lab-02-2026-it26100408
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408$ cd lab-02-2026-it26100408
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408/lab-02-2026-it26100408$ vi hello_world.c
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408/lab-02-2026-it26100408$ nano hello_world.c
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408/lab-02-2026-it26100408$ gcc hellow
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot find hellow: No such file or directory
collect2: error: ld returned 1 exit status
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408/lab-02-2026-it26100408$ gcc hello_world.c -o hello_world
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408/lab-02-2026-it26100408$ ./hello_world
./hello_world: command not found
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408/lab-02-2026-it26100408$ ./hello_world
Hello! Welcome to SLIIT
First line
This is the second line
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408/lab-02-2026-it26100408$ git add hello_world.c
vihangam@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408/lab-02-2026-it26100408$ git commit -m "lab_activity1"
Author identity unknown
```

```

vihanga@UbuntuVM:~/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408$ nano -/
vihanga@UbuntuVM: $ pwd
/home/vihanga/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408$ source
vihanga@UbuntuVM: $
vihanga@UbuntuVM: $ git clone https://github.com/it26100408/PM-2-LAB.git
Cloning into 'PM-2-LAB'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 8 (delta 0), reused 8 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), done.
vihanga@UbuntuVM: $ ls
PM-2-LAB  hello_world  hello_world.c
vihanga@UbuntuVM: $ mkdir problem3
vihanga@UbuntuVM: $ cd problem3
vihanga@UbuntuVM: $ nano pattern.c
vihanga@UbuntuVM: $ gcc pattern.c -o pattern.c
gcc: fatal error: input file 'pattern.c' is the same as output file
compilation terminated.
vihanga@UbuntuVM: $ gcc pattern.c -o pattern
vihanga@UbuntuVM: $ ./pattern
  x
 x x
x  x
  x x
 x x
  x x
 x x
  x x
  x
vihanga@UbuntuVM: $ git add pattern.c

```

Activity 2: Calculating Missing Heights

In an experiment, the heights of 5 people were recorded and the average height was computed. Due to negligence, the height of two people has been lost. The lab assistant distinctly remembers that the two missing heights were almost the same. Develop a C program to input the heights of the known three people and the calculated average. Find and print the missing heights.

Instructions

1. Open the Terminal using the Applications Menu or the keyboard shortcut (Ctrl + Alt + T).
2. Navigate to the working directory you created on the Desktop.
3. Create a new C file named `missing_heights.c`.
4. Write the C program as described in Problem 2.
5. Compile and run the program to ensure it works correctly.

Code:

PM-2-LAB / missing_heightsfinal.c 

 it26100408 missing_heightsfinal

Code Blame 24 lines (14 loc) · 528 Bytes

```
1
2  #include <stdio.h>
3
4  int main() {
5      float height1, height2, height3, average, total, missing1, missing2;
6
7      printf("Enter the heights of three known people: ");
8      scanf("%f %f %f", &height1, &height2, &height3);
9
10     printf("Enter the average height: ");
11     scanf("%f", &average);
12
13     total = average * 5;
14
15     missing1 = (total - (height1 + height2 + height3))/2;
16     missing2 = missing1;
17
18     printf("\nMissing height 1 = %.2f", missing1);
19     printf("\nMissing height 2 = %.2f", missing2);
20
21     return 0;
22 }
```


Out put:

```
vihangar@UbuntuVM: ~/Desktop/IT26100408 $ cd PM-02-LAB
bash: cd: PM-02-LAB: No such file or directory
vihangar@UbuntuVM: ~/Desktop/IT26100408 $ mkdir PM-02-LAB
vihangar@UbuntuVM: ~/Desktop/IT26100408 $ cd PM-02-LAB
vihangar@UbuntuVM: ~/Desktop/IT26100408/PM-02-LAB $ mkdir lab-02-2026-it26100408
vihangar@UbuntuVM: ~/Desktop/IT26100408/PM-02-LAB $ cd lab-02-2026-it26100408
vihangar@UbuntuVM: ~/Desktop/IT26100408/PM-02-LAB/lab-02-2026-it26100408 $ nano missing_heights.c
vihangar@UbuntuVM: ~/Desktop/IT26100408/PM-02-LAB/lab-02-2026-it26100408 $ gcc missing_heights.c -missing_heights
gcc: error: unrecognized command-line option '-missing_heights'
vihangar@UbuntuVM: ~/Desktop/IT26100408/PM-02-LAB/lab-02-2026-it26100408 $ gcc missing_heights.c -o missing_heights
vihangar@UbuntuVM: ~/Desktop/IT26100408/PM-02-LAB/lab-02-2026-it26100408 $ ./missing_heights.c
bash: ./: Is a directory
vihangar@UbuntuVM: ~/Desktop/IT26100408/PM-02-LAB/lab-02-2026-it26100408 $ ./missing_heights
bash: ./: Is a directory
vihangar@UbuntuVM: ~/Desktop/IT26100408/PM-02-LAB/lab-02-2026-it26100408 $ ./missing_heights
Enter the heights of three known people: 156.6
165.8
161.9
Enter the average height: 161.43

Missing height 1 = 161.42
Missing height 2 = 161.42vihangar@UbuntuVM: ~/Desktop/IT26100408/PM-02-LAB/lab-02-2026-it26100408 $
```

Activity 3: Calculating Fence Dimensions

Write a C program to determine the width and length of a rectangular fence where the perimeter is known. If the width of the fence is $\frac{3}{4}$ the size of the length of the fence. The program should take the perimeter as input and calculate the width and length.

Instructions

1. Open the Terminal using the Applications Menu or the keyboard shortcut (Ctrl + Alt + T).
2. Navigate to the working directory you created on the Desktop.
3. Create a new C file named `fence_dimensions.c`.
4. Write the C program as described in Problem 3.
5. Compile and run the program to ensure it works correctly.

Code:

PM-2-LAB / lab-02-2026-it26100408 / lab-02-2026-it26100408 / problem3 / fence_dimensions.c 

 it26100408 Update fence dimension calculation formula

Code Blame 16 lines (11 loc) · 446 Bytes

```
1  #include <stdio.h>
2
3  int main(){
4      float perimeter, length,width;
5
6      printf("Enter the perimeter of the fence: ");
7      scanf("%f", &perimeter);
8
9      length = perimeter / 3.5;
10     width = 0.75 * length;
11
12     printf("Length of the fence: %.2f\n", length);
13     printf("Width of the fence: %.2f\n", width);
14
15     return 0;
16 }
```

Out put:

```
/home/vihanga/Desktop/IT26100408/PM-2-LAB/lab-02-2026-it26100408/lab-02-2026-it26100408/problem3
vihanga@UbuntuVM: $ nano fence_dimensions.c
vihanga@UbuntuVM: $ gcc fence_dimensions.c -o fence_dimensions
vihanga@UbuntuVM: $ ./fence
bash: ./fence: No such file or directory
vihanga@UbuntuVM: $ ./fence_dimensions
Enter the perimeter of the fence: 100
Length of the fence: 20.57
Width of the fence: 21.43
vihanga@UbuntuVM: $ git add fence_dimensions.c
vihanga@UbuntuVM: $ git -m "Completed problem 4: Fence dimensions"
unknown option: -m
usage: git [-v | --version] [-h | --help] [-C <path>] [-c <name>=<value>]
    [--exec-path[=<path>]] [--html-path] [--man-path] [--info-path]
    [-p | --paginate | -P | --no-pager] [--no-replace-objects] [--no-lazy-fetch]
    [--no-optionallocks] [--no-advice] [--bare] [--git-dir=<path>]
    [--work-tree=<path>] [--namespace=<name>] [--config-env=<name>=<envvar>]
    <command> [<args>]
vihanga@UbuntuVM: $ git push
Username for 'https://github.com': it26100408@my.sliit.lk
Password for 'https://it26100408@my.sliit.lk@github.com':
Everything up-to-date
vihanga@UbuntuVM: $ git commit -m "Completed problem 4: Fence dimensions"
[main 19abb5] Completed problem 4: Fence dimensions
1 file changed, 20 insertions(+)
create mode 100644 lab-02-2026-it26100408/lab-02-2026-it26100408/problem3/fence_dimensions.c
vihanga@UbuntuVM: $ git push
Username for 'https://github.com': it26100408@my.sliit.lk
Password for 'https://it26100408@my.sliit.lk@github.com':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 822 bytes | 822.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/it26100408/PM-2-LAB.git
410ed44..19abb5 main -> main
vihanga@UbuntuVM: $
```

Thank You!